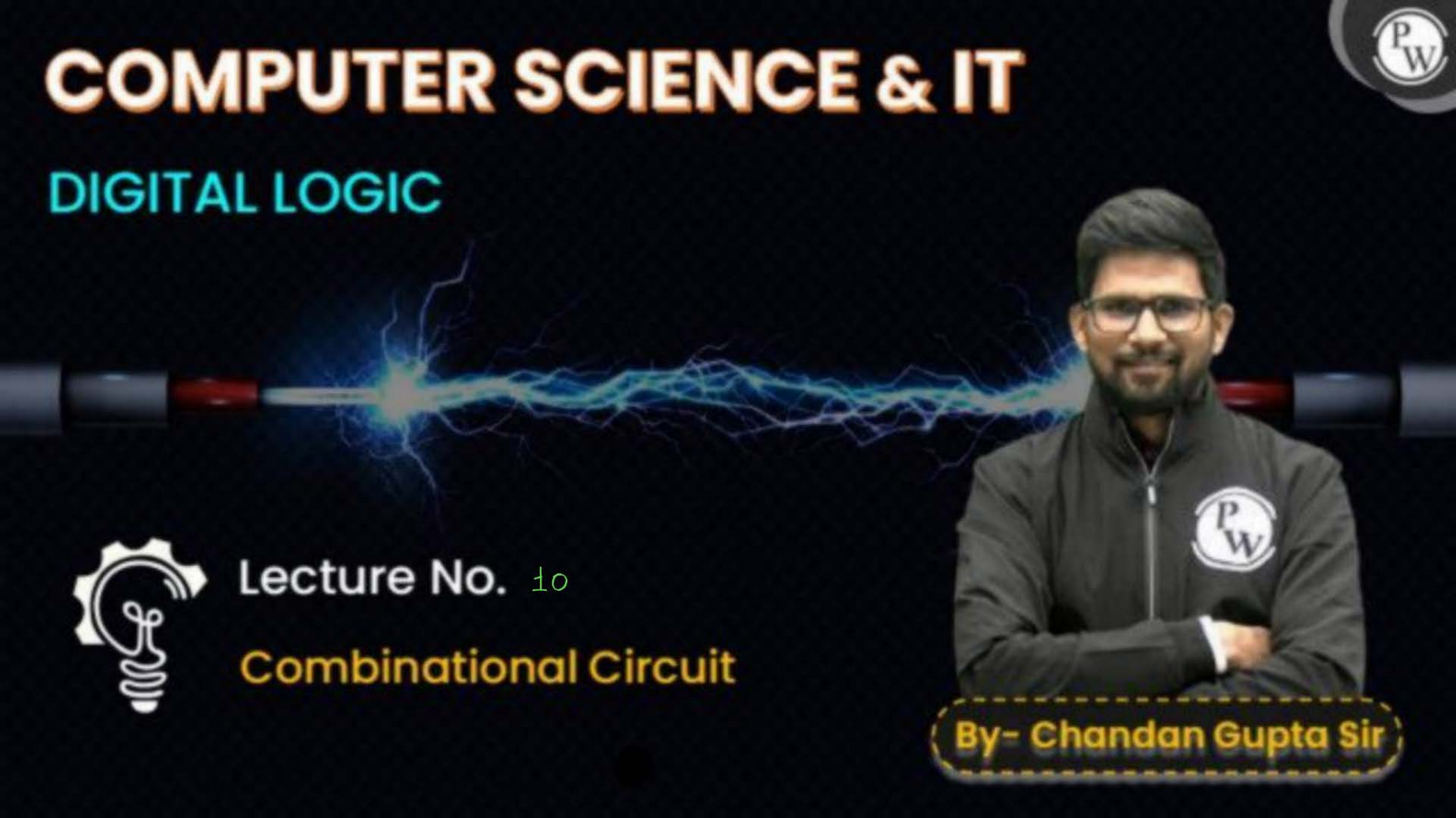
COMPUTER SCIENCE & IT

DIGITAL LOGIC

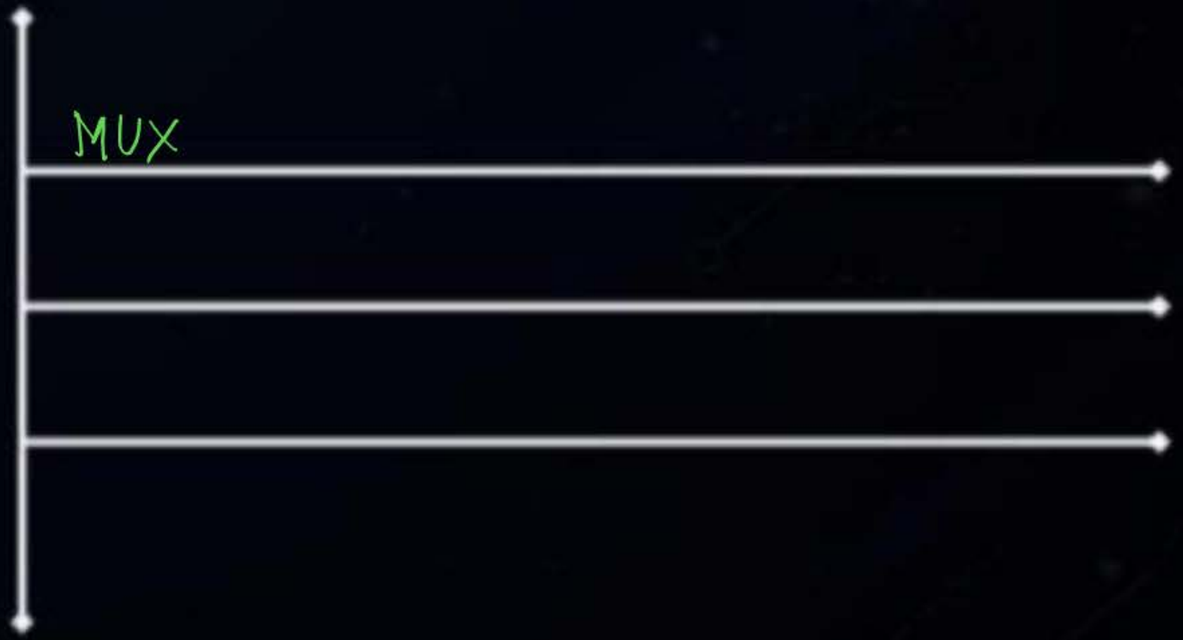


Lecture No. 10

Combinational Circuit

By- Chandan Gupta Sir

Recap of Previous Lecture





MUX Cont.

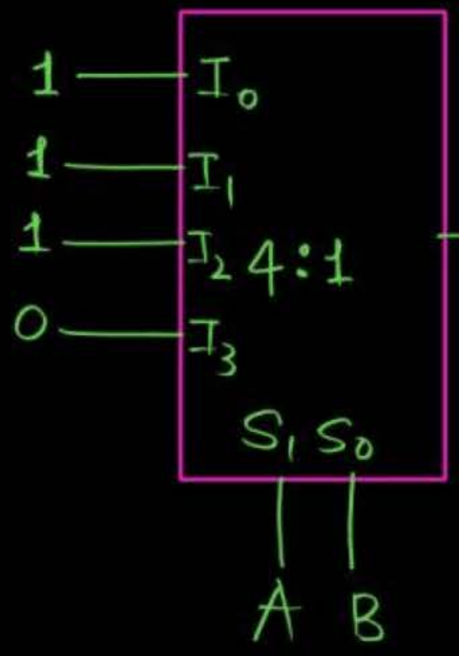
Implementation of Boolean function using MUX

Q. $f(A, B) = \Sigma(0, 1, 2)$ → implement this logical function

(i) using 4 : 1 MUX

(ii) using 2 : 1 MUX

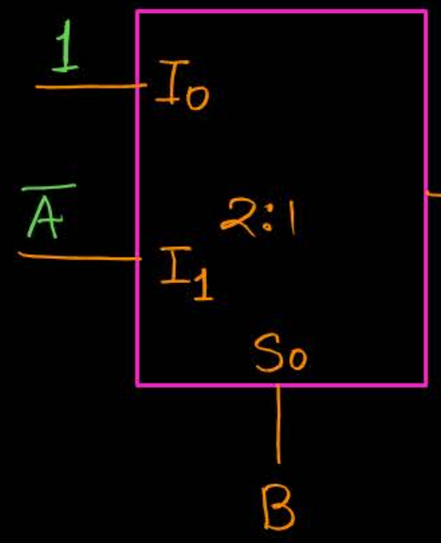
(i).



$$f(A,B) = \bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 1 + A\bar{B}\cdot 1 + AB\cdot 0$$

$$= \underset{00}{\bar{A}\bar{B}} + \underset{01}{\bar{A}B} + \underset{10}{A\bar{B}}$$

(11).



$$= \overline{B} \cdot 1 + B \cdot \overline{A}$$

$$= \left(\overline{B} + \overline{A}\right) = \overline{A \cdot B}$$

		I_0	I_1
0	$\overline{A}$	0	1
1	A	2	3
		1	$\overline{A}$

2^2 2^1 2^0
A B C

$\overline{C} \longrightarrow 0, 2, 4, 6$

$C \longrightarrow 1, 3, 5, 7$

$\overline{B} \longrightarrow 0, 1, 4, 5$

$B \longrightarrow 2, 3, 6, 7$

$\overline{A} \longrightarrow 0, 1, 2, 3$

$A \longrightarrow 4, 5, 6, 7$

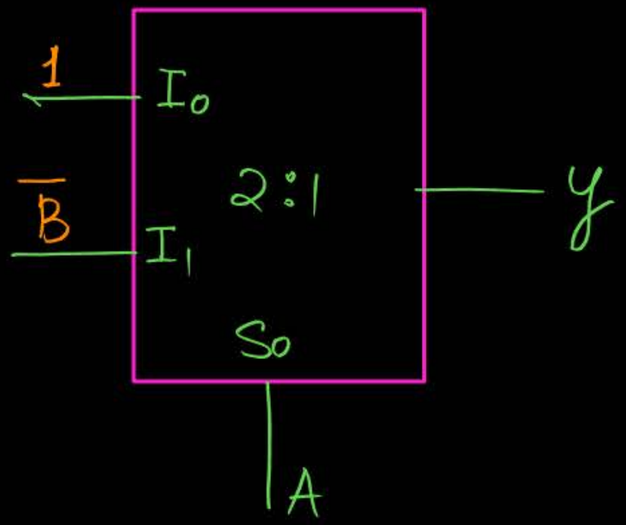
2^1 2^0
A B

$\overline{B} \rightarrow 0, 2$

$B \rightarrow 1, 3$

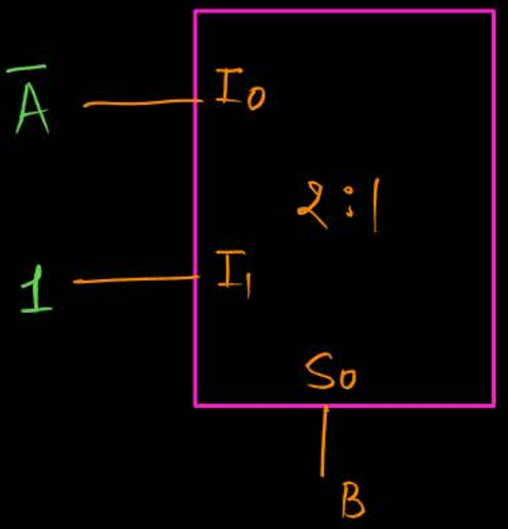
$\overline{A} \rightarrow 0, 1$

$A \rightarrow 2, 3$

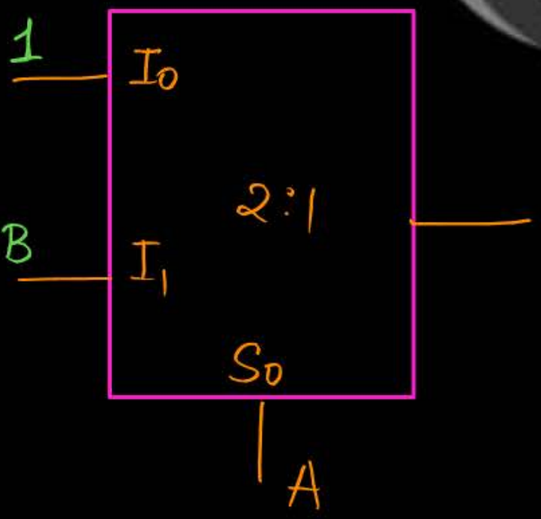


	I_0	I_1
$\bar{B}$	0	2
B	1	3
	1	$\bar{B}$

- $f(A,B) = \Sigma(0,1,3)$



	I_0	I_1
$\bar{A}$	0	1
A	2	3
	$\bar{A}$	1

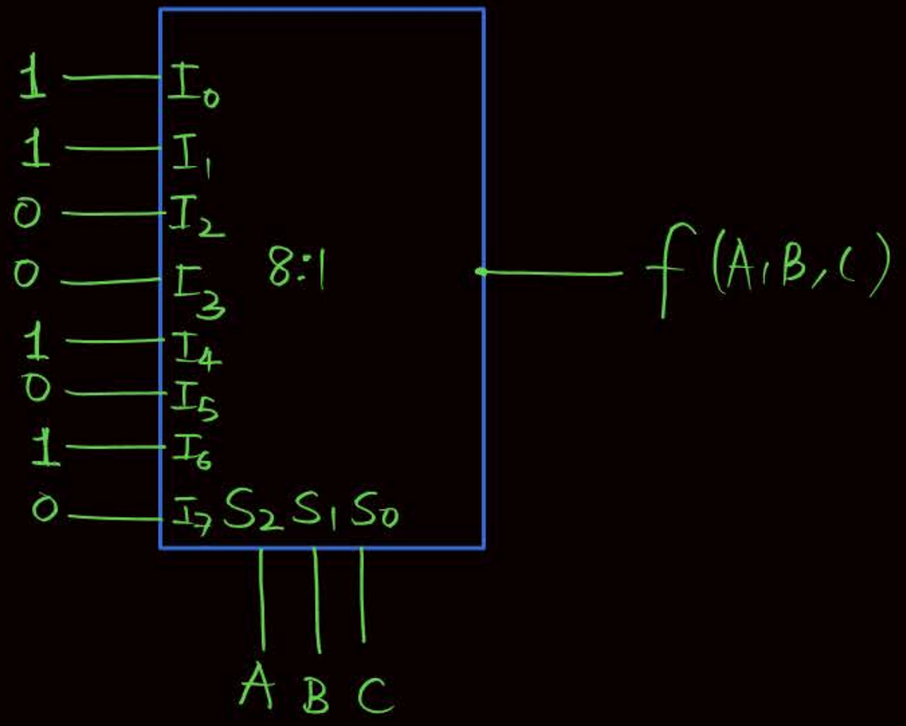


	I_0	I_1
$\bar{B}$	0	2
B	1	3
	1	B

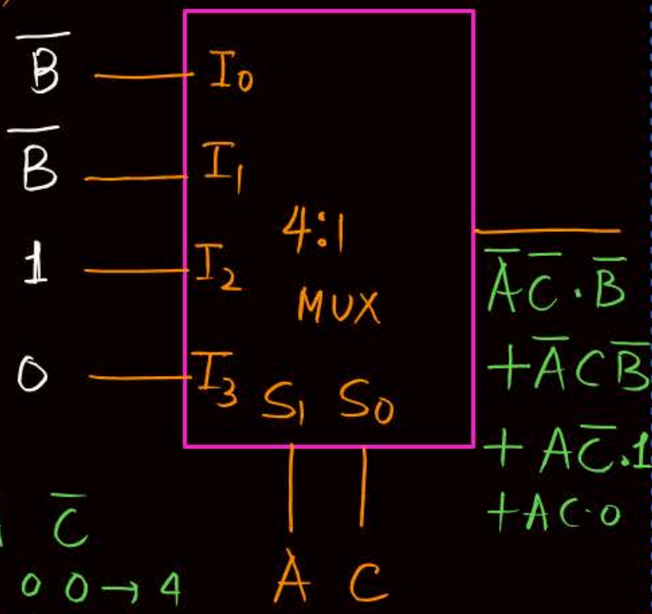
Q. $f(A,B,C) = \Sigma(0,1,4,6)$

(i) 8:1 MUX

(ii) 4:1 MUX



(ii)

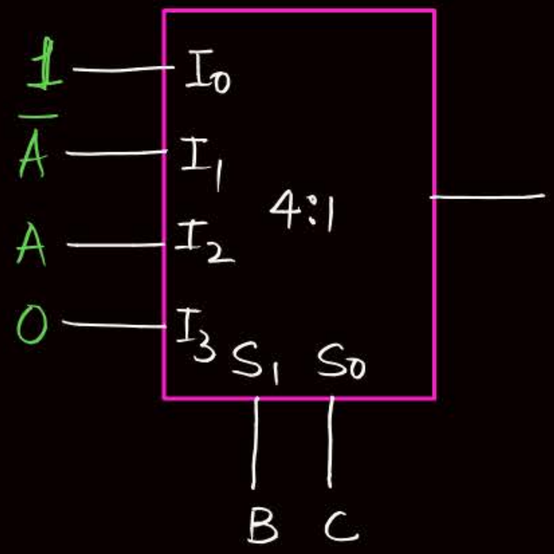


A $\bar{C}$

1 0 0 → 4

1 1 0 → 6

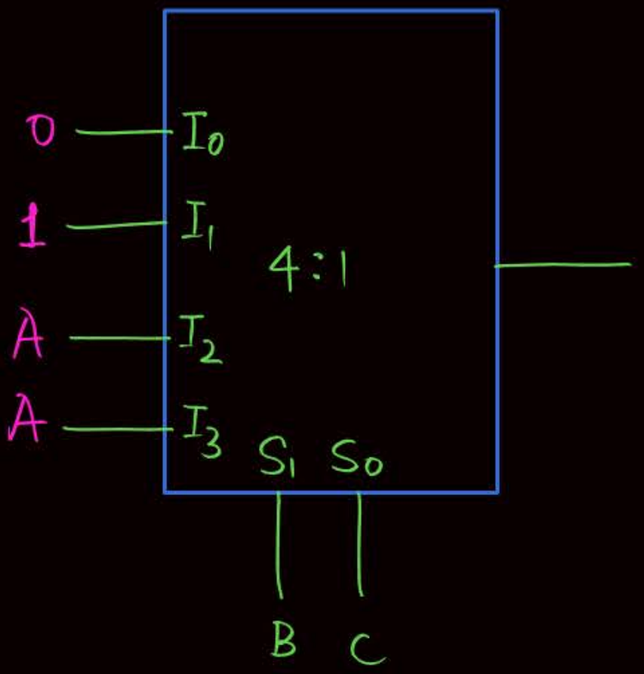
	I_0	I_1	I_2	I_3
$\bar{B}$	(0)	(1)	(4)	5
$\bar{B}$	2	3	(6)	7
	$\bar{B}$	$\bar{B}$	1	0



	I_0	I_1	I_2	I_3
$\bar{A}$	(0)	(1)	2	3
A	(4)	5	(6)	7
	1	$\bar{A}$	A	0

- $f(A,B,C) = \Sigma(1,5,6,7)$

  using 4:1 MUX and B, C as select line.



	I_0	I_1	I_2	I_3
$\bar{A}$	0	(1)	2	3
A	4	(5)	(6)	(7)
	0	1	$\bar{A}$	$\bar{A}$

- 4 : 1 MUX can be used to implement

All 2-variable function

Some of the 3-variable function.

- 4 : 1 MUX + 1 NOT GATE can be used to implement
  - All 3-variable function.

- 8 : 1 MUX can be used to implement

All 3-variable function

Some of the 4-variable function.

- 8 : 1 MUX + 1 NOT GATE can be used to implement
  - All 4 – variable function.

- $2^n : 1$ MUX can be used to implement

All n-variable function

Some of (n+1) variable function.

- $2^n : 1$ MUX + 1 NOT GATE can be used to implement
  - All $(n+1)$ variable function.

MSQ

Q. We can implement all n-variable function using.

a. $2^n : 1$ MUX

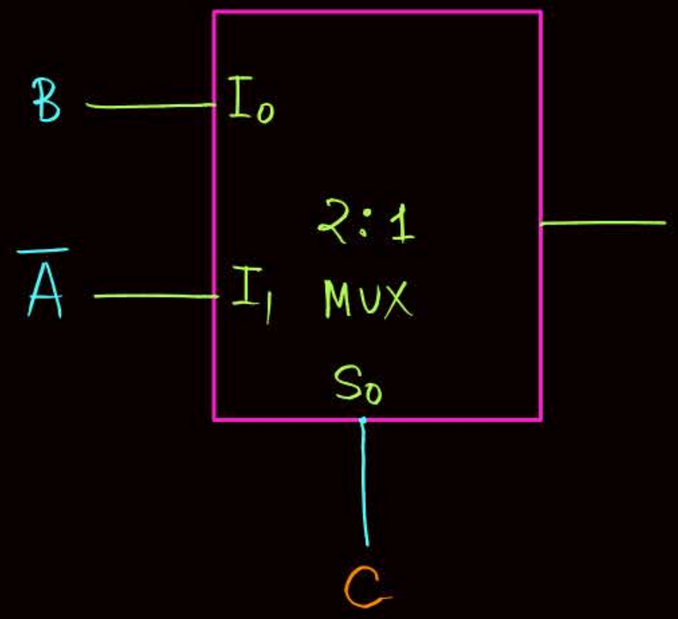
b. $2^n : 1$ MUX + 1 NOT gate

c. $2^{n-1} : 1$ MUX + 1 NOT gate

d. $2^{n+1} : 1$ MUX + 1 NOT gate.

#Q. $f(A,B,C) = \Sigma(1,2,3,6)$

using (2:1) MUX



0 0 0 → 0
0 0 1 → 1

0 1 0
0 1 1

$f(A,B,0) = \Sigma(2,6) = \bar{A}B\cdot 1 + AB\cdot 1$

$f(A,B,1) = \Sigma(1,3) = \bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 1$

$= B \Rightarrow I_0 = B$

$= \bar{A} \rightarrow I_1 = \bar{A}$

		I_0	I_1
00	$\bar{A}\bar{B}$	0	(1)
01	$\bar{A}B$	(2)	(3)
10	$A\bar{B}$	4	5
11	AB	(6)	7
		$\bar{A}B + AB$	$\bar{A}\bar{B} + \bar{A}B$
		$= B$	$= \bar{A}$

- $f(A,B,C,D) = (A+\overline{C})\cdot(\overline{B}+D)$ using 4:1 MUX and B & C as select line.

$= \pi(2,3,4,6,7,12,14)$

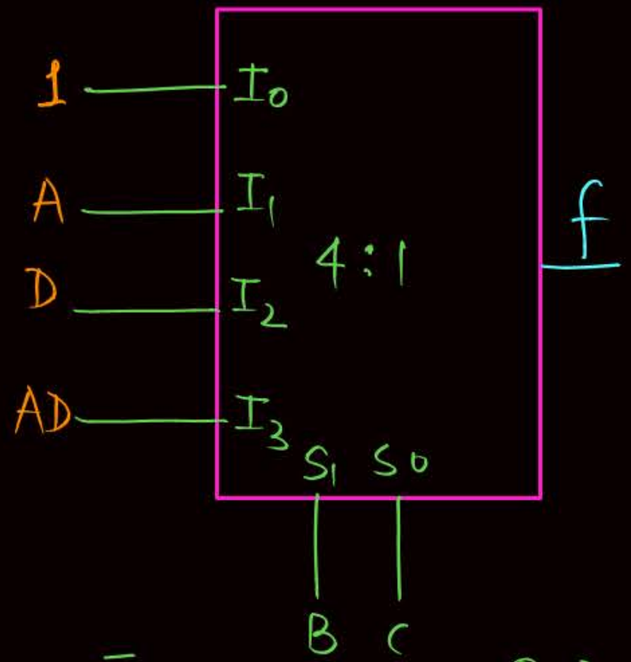
$= \Sigma(0,1,5,8,9,10,11,13,15)$

$(A+\overline{C})$

0 0 | 0 → 2
0 0 | 1 → 3
0 1 | 0 → 6
0 1 | 1 → 7

$(\overline{B}+D)$

0 | 0 0 → 4
0 | 1 1 → 6
1 | 1 0 → 12
1 | 1 1 → 14



	I_0	I_1	I_2	I_3
$\overline{A}\,\overline{D}$	0	2	4	6
$\overline{A}D$	1	3	5	7
$A\overline{D}$	8	10	12	14
AD	9	11	13	15
	1	A	D	AD

$(A\overline{D}+AD)$

$\overline{A}$ D

0 0 0 | 1
0 0 | 1 1
0 | 0 | 1
0 | 1 | 1

0 0 0 0
0 0 | 1 0
0 | 1 0 0
0 | 1 1 0

A $\overline{D}$

1 0 0 0

$f(A, 0, 0, D) = 1 \longrightarrow I_0 = 1$

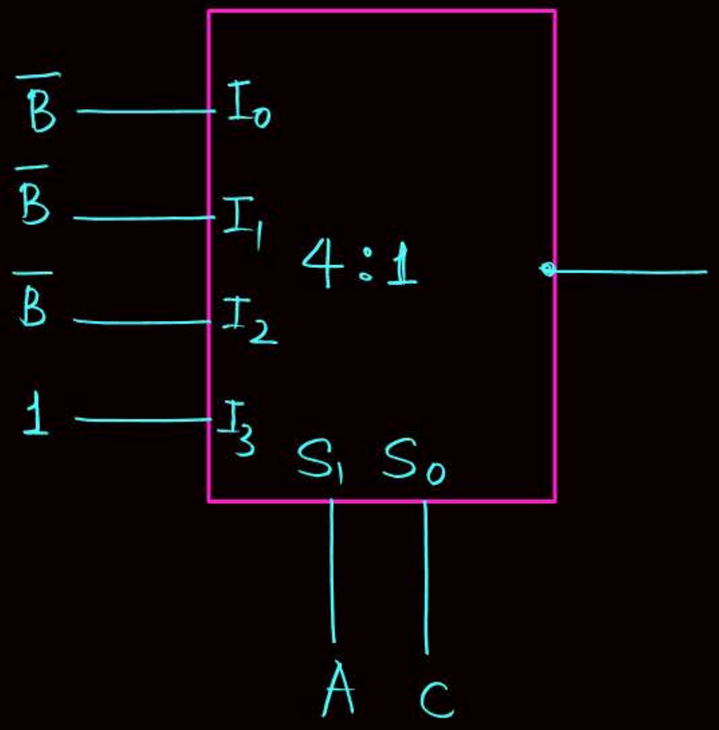
$f(A, 0, 1, D) = A \longrightarrow I_1 = A$

$f(A, 1, 0, D) = D \longrightarrow I_2 = D$

$f(A, 1, 1, D) = AD \longrightarrow I_3 = AD$

#Q. $f(A,B,C) = \overline{\bar{A}B + B\bar{C}} + AC$

implement it using 4:1 MUX and A C as select line.



$f(0,B,0) = \bar{B} \longrightarrow I_0 = \bar{B}$

$f(0,B,1) = \bar{B} \longrightarrow I_1 = \bar{B}$

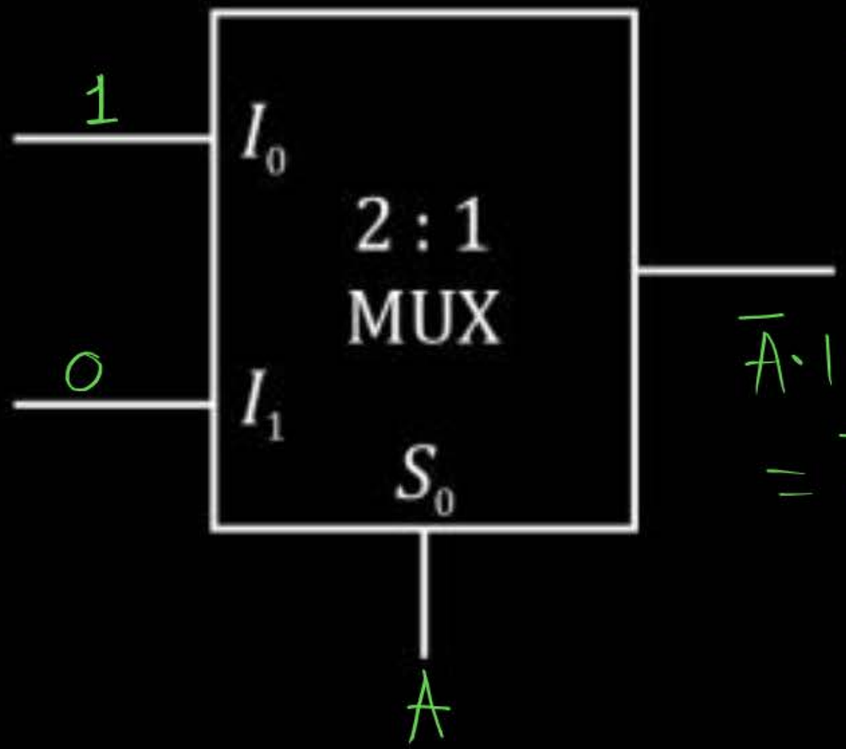
$f(1,B,0) = \bar{B} \longrightarrow I_2 = \bar{B}$

$f(1,B,1) = 1 \longrightarrow I_3 = 1$

MUX as Universal Circuit

- NOT GATE : $f(A) = \bar{A}$, $f(0) = 1 \longrightarrow I_0 = 1$

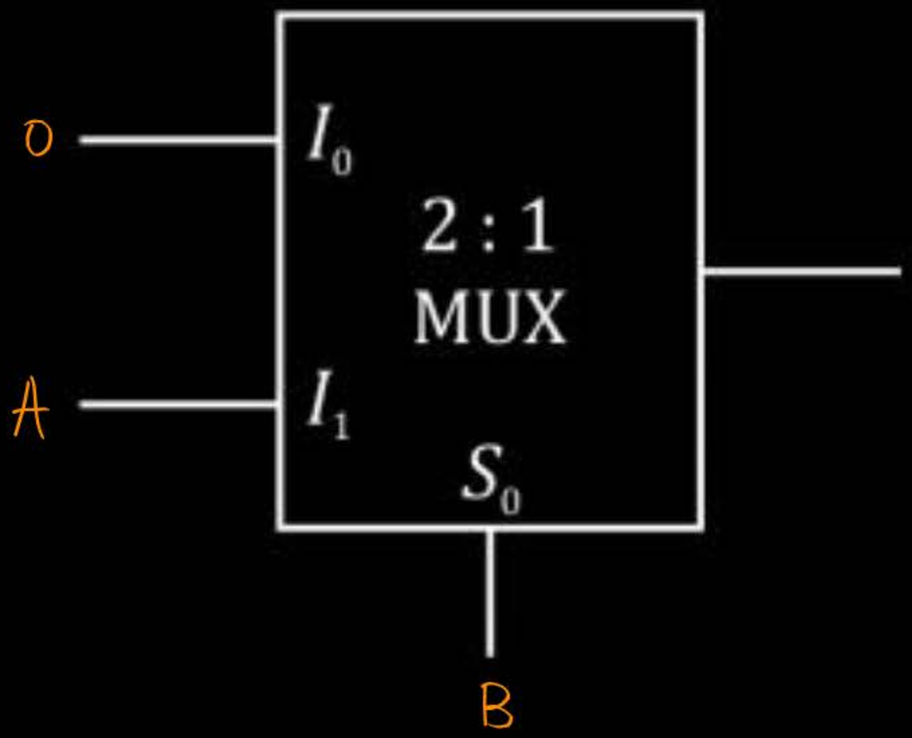
  $f(1) = 0 \longrightarrow I_1 = 0$



$\bar{A} \cdot 1 + A \cdot 0$

$= \bar{A}$

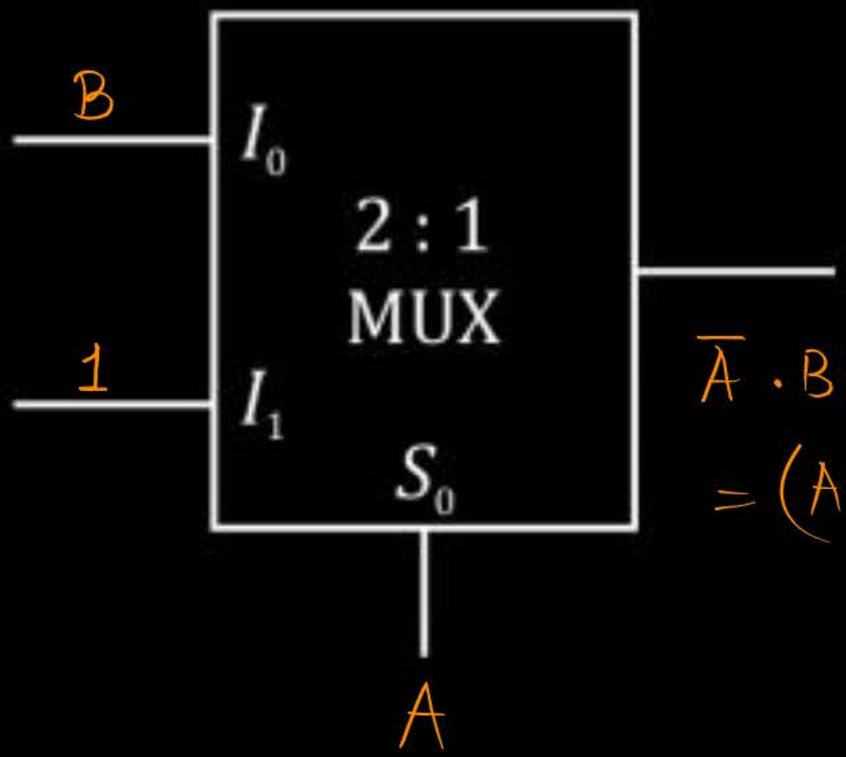
- AND GATE : $f(A, B) = A.B = \Sigma(3)$



$f(A,0) = A \cdot 0 = 0 \longrightarrow I_0 = 0$

$f(A,1) = A \cdot 1 = A \longrightarrow I_1 = A$

- OR GATE : $f(A, B) = (A + B) = \Sigma(1,2,3)$



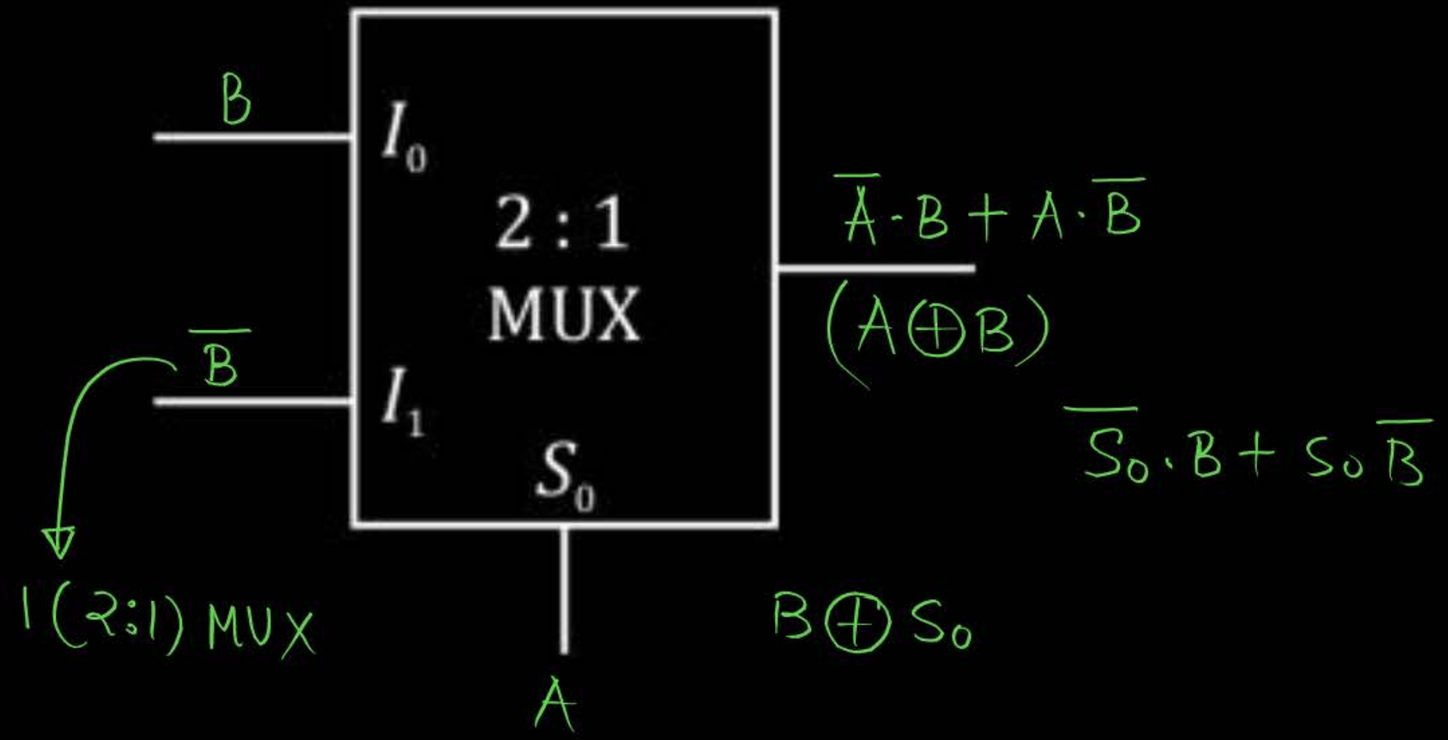
$f(0, B) = 0 + B = B \longrightarrow I_0 = B$

$f(1, B) = 1 + B = 1 \longrightarrow I_1 = 1$

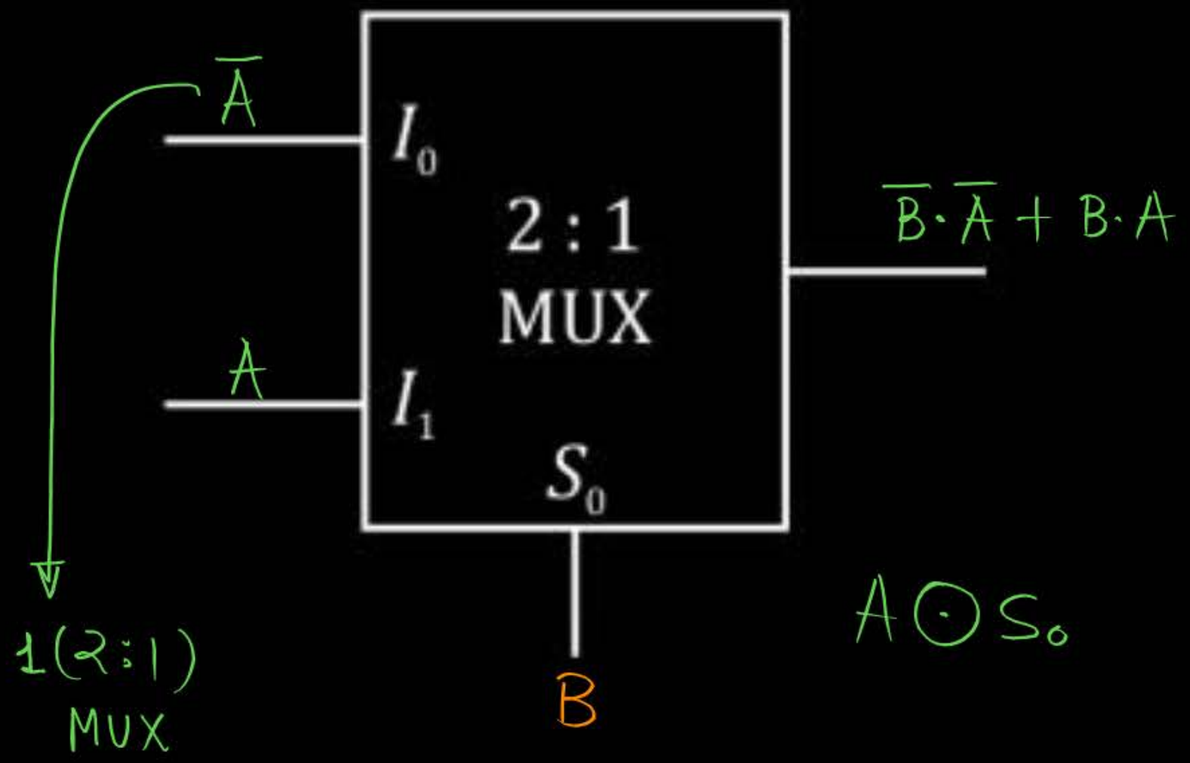
$\overline{A} \cdot B + A \cdot 1$

$= (A + B)$

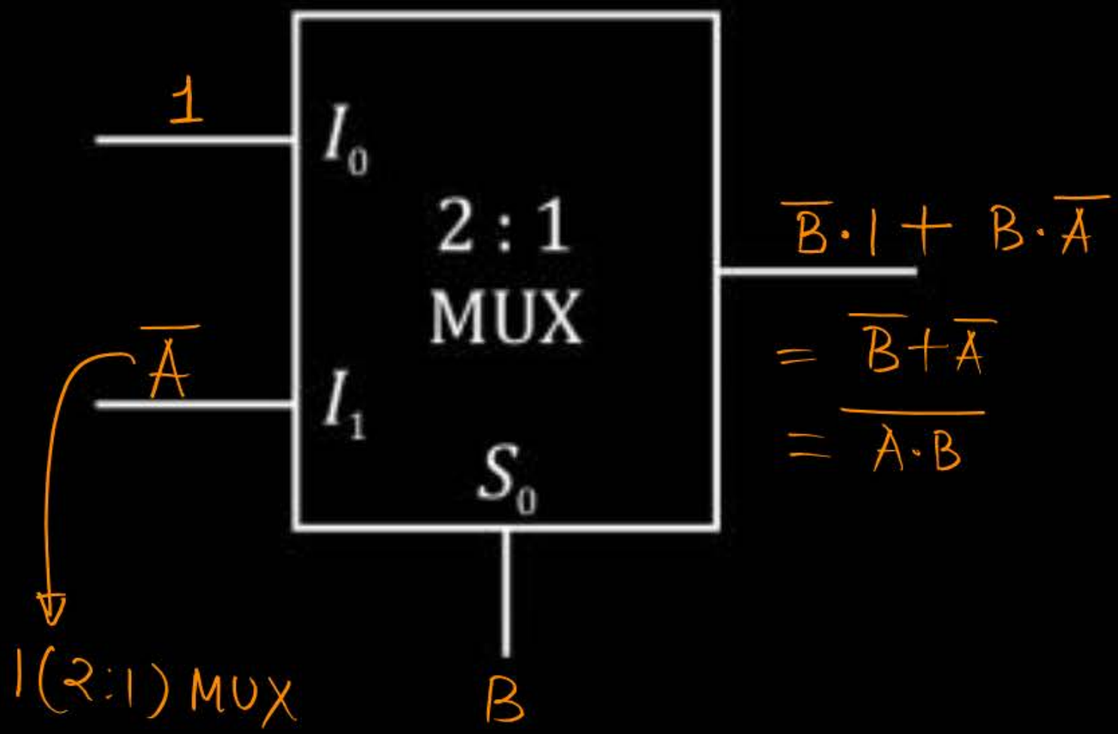
- XOR GATE : $f(A,B) = (A \oplus B) = \Sigma(1,2)$



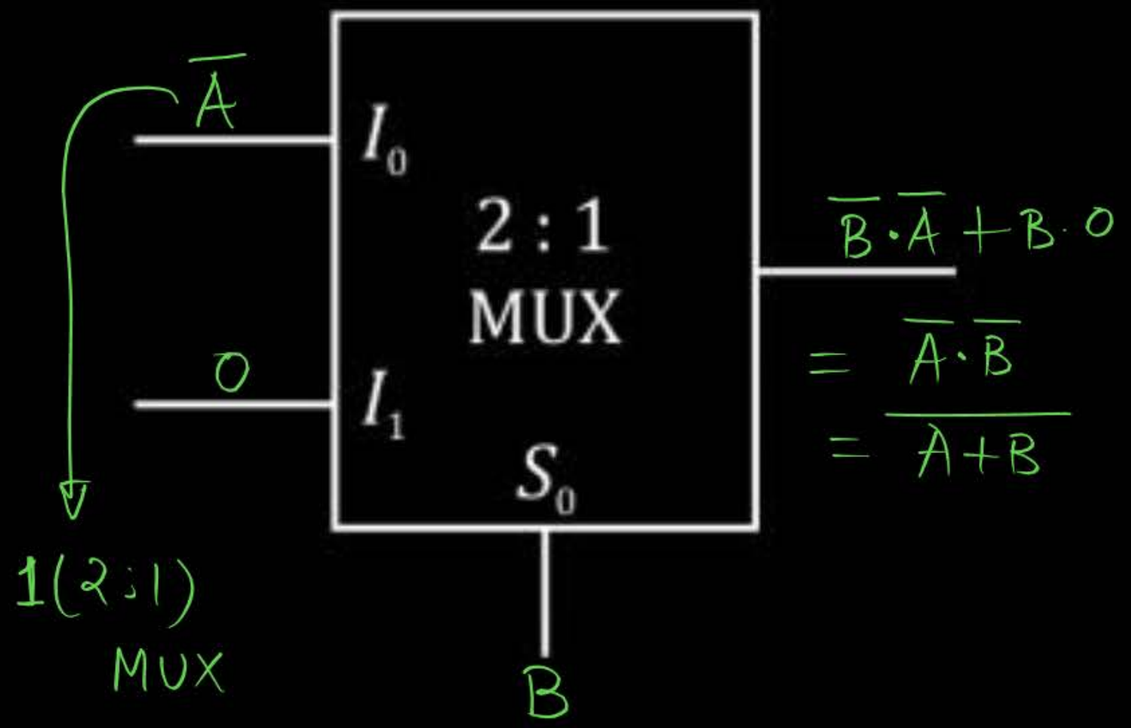
- XNOR GATE : $f(A,B) = (A \odot B) = \Sigma(0,3)$



- NAND GATE : $f(A,B) = \overline{A \cdot B}$ $= \Sigma(0,1,2)$



- NOR GATE : $f(A,B) = \overline{A+B} = \Sigma(0)$

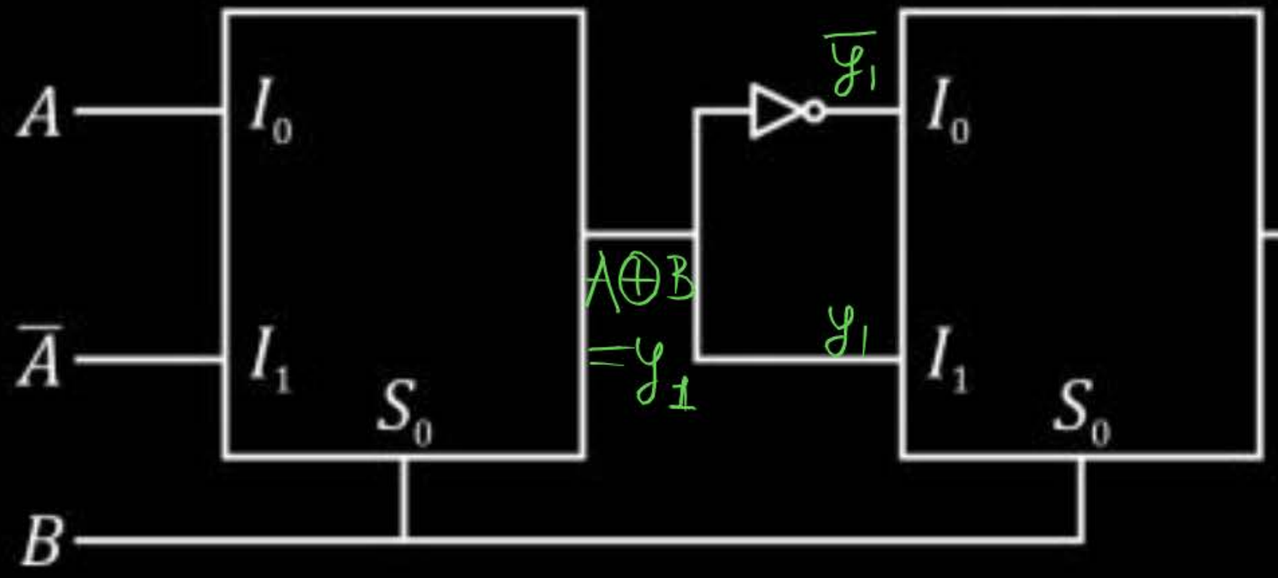


- IMP Notes :

- To implement basic gates [NOT, OR, AND] we require only one (2:1) MUX.
- To implement Non-basic gates [XOR, XNOR, NAND, NOR] we require two (2:1) MUX.

Questions On MUX

A digital circuit is as given below :



Output y is

(a) $A \oplus B$

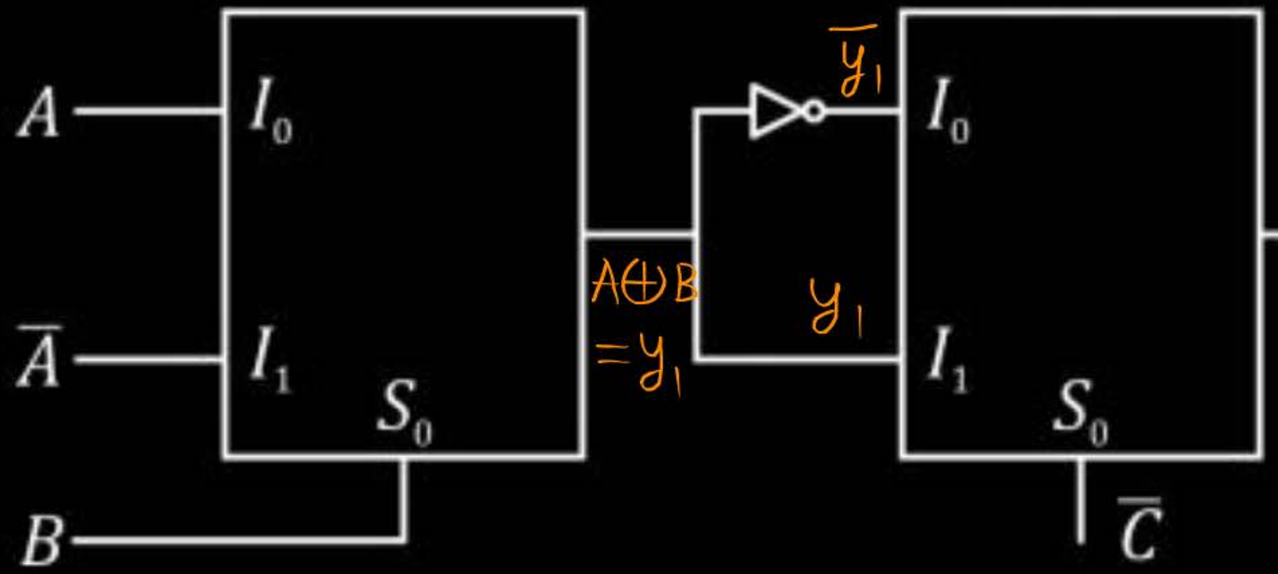
(b) $A \odot B$

(c) $C \,.\, A$

(d) $\bar{A}$

Question 2

A digital circuit is as given below :



Output y is

(a) $A \oplus B \oplus C$

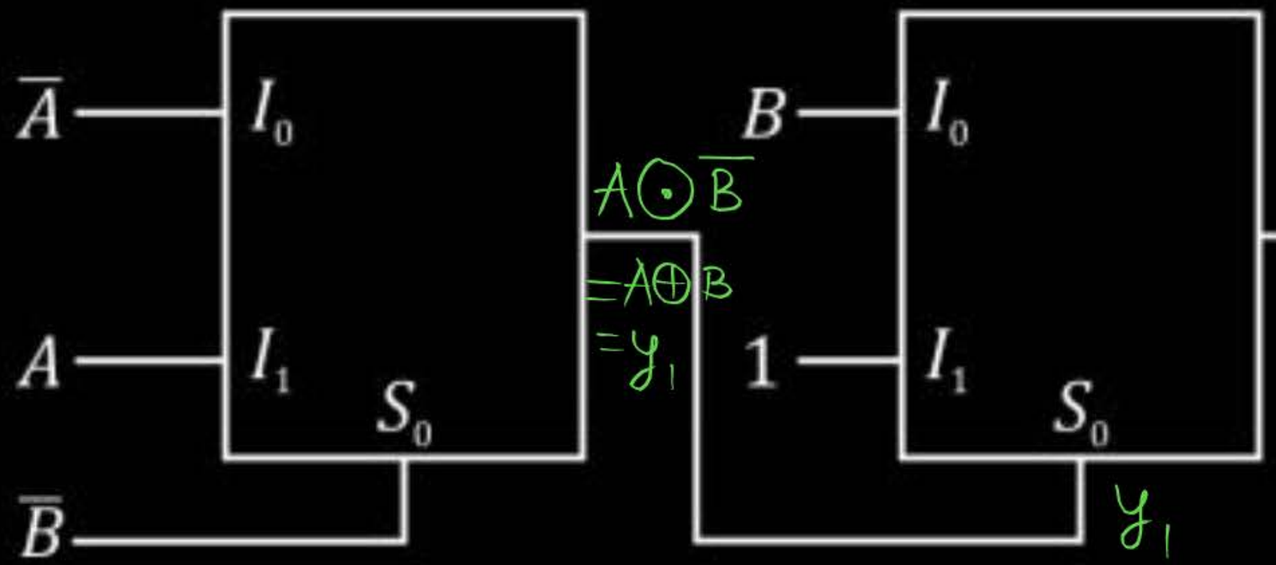
(b) $A \oplus B \odot C$

(c) $A \oplus C$

(d) $A \oplus B$

Question 3

A digital circuit is as given below :



$y = (\overline{y_1} \cdot B) + y_1 \cdot 1$

$= y_1 + B$

$= A \oplus B + B$

$= \overline{A}B + A\overline{B} + B$

$= B + (A\overline{B})$

$= (B + A) \cdot (B + \overline{B})$

$= (A + B)$

Output y is

(a) $A \oplus B$

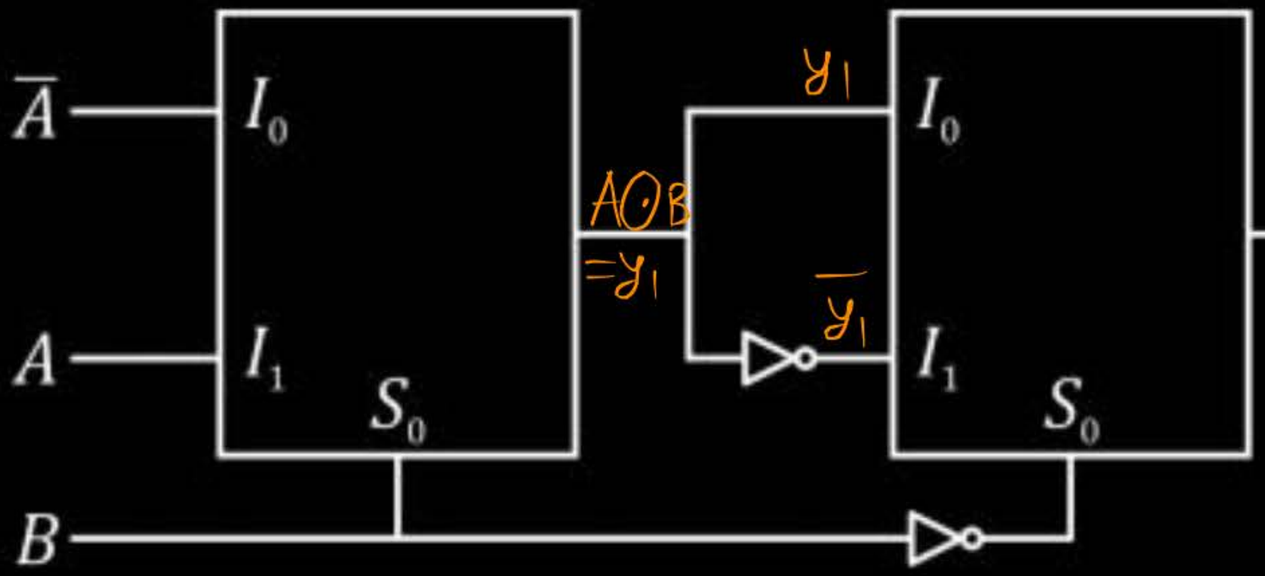
(b) $A \odot B$

(c) $(A + B)$

(d) $A \,.\, B$

Question 4

A digital circuit is as given below :



$y = y_1 \oplus \overline{B}$

$= A \odot B \oplus \overline{B}$

$= A \odot 1$

$= A$

Output y is

(a) B

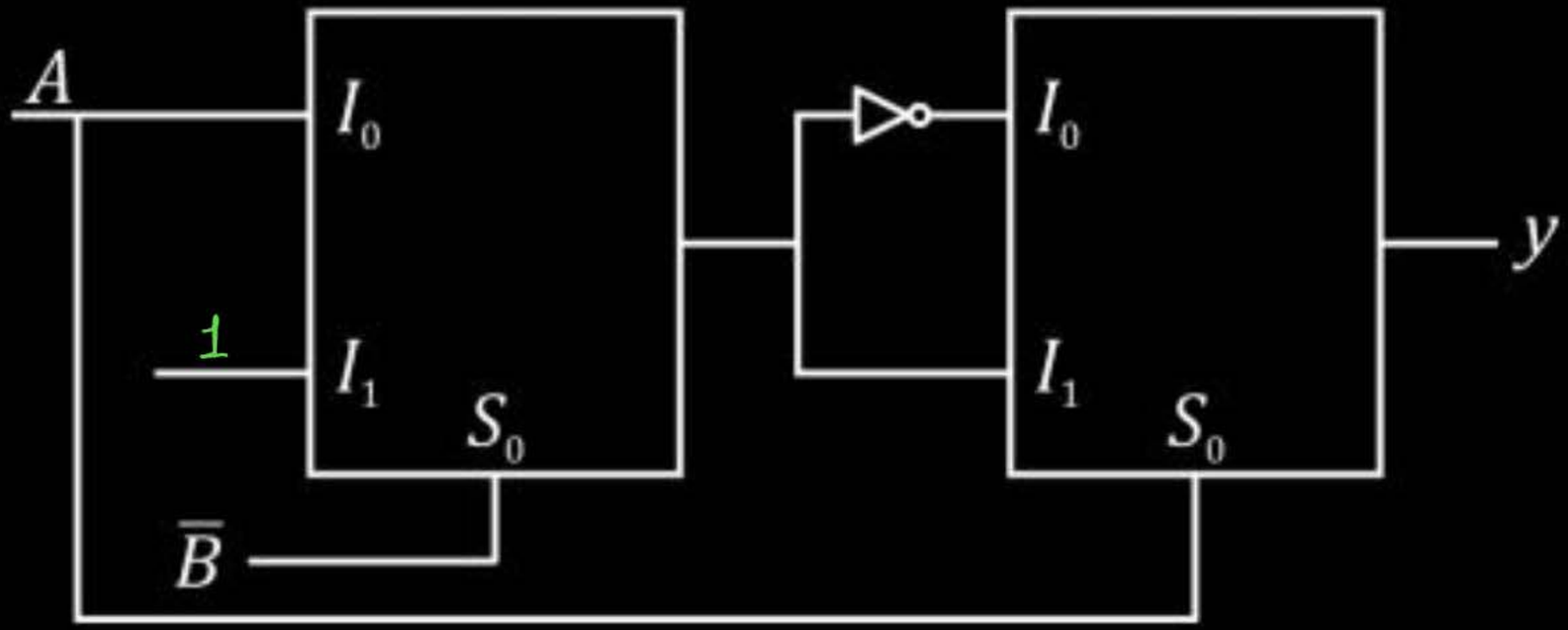
(b) A

(c) $(A + B)$

(d) $A \,.\, B$

Question 5

A digital circuit is as given below :



H.W.

The output y is

(a) $A \odot B$

(b) $\bar{A}B$

(c) $(A + B)$

(d) $A\bar{B}$



2 Minute Summary

MUX

Thank you

GW

Soldiers !

